UBC APPLIED SCIENCE CO-OP STUDENT SUMMARY

Student Name (Last, Preferred First):	Tian, Brian
Academic Terms Completed by Position Start Date:	10
Work Terms Completed by Position Start Date:	2
Available up to (4/8/12/16 months):	4/8/12
Discipline:	Electrical Engineering
Eligible for SWPP* Funding (y/n): *Students must be a Canadian citizen, permanent resident or hold refugee status to be eligible for SWPP funding	n
Driver's Licence (Optional) *Specify License Class/Type	n/a
Citizenship Status	International student with a valid Canadian co-op work permit; currently residing in Canada

Date Completed: (mm/dd/yyyy) July 15th, 2026

Student Summary information is self-reported by the student.

Note: Please ensure this is uploaded to PD Portal as a PDF.

Brian Tian
5868 Agronomy Road, Vancouver, BC, V6T 0B5
604-312-3703 | jxtlx98@163.com | linkedin.com/in/lingxiao-tian
Citizenship status: International student | Valid co-op work permit

July 16th, 2026

Hiring manager
Mazdis Innovations Inc.
6163 University Blvd, Vancouver, BC V6T 1Z1
Re: Electrical & Software Systems Engineer (Robotics & Automation)

Dear Hiring Manager:

I am applying for the Electrical & Software Systems Engineer at Mazdis Innovations. As a fourth-year Electrical Engineering student at the University of British Columbia, I have developed hands-on experience in embedded programming, hardware-software integration, and system debugging through robotics and control projects. I am drawn to MAZDIS's mission of developing next-generation robotic storage systems that integrate embedded control and intelligent automation, and my experience with Embedded C, ESP32, Arduino, Python and hands-on system testing aligns well with the role's embedded systems responsibilities. I am available for a 4/8/12-month work term and would be committed to contributing consistently to your team.

In a laser-projector project, I developed Embedded C firmware that used a timer-driven interrupt service routine to execute a PID control loop under a specified ISR-utilization requirement. When our team initially lacked a reliable way to measure the ISR execution time, I proposed driving a GPIO pin high at the beginning of the ISR and low at the end, allowing us to measure the pulse width and interrupt period with an oscilloscope. Using those measurements together with the MCU clock frequency and timer prescaler, I selected an interrupt interval that kept ISR utilization within the required 50–70% range and supported stable operation. I also translated the projector's operating sequence into a finite-state machine, defining the logic and transitions between each mode. This project reflects how I approach embedded development: learning unfamiliar hardware quickly, validating firmware directly on devices, and refining designs through measured results.

During my co-op at Convergent Manufacturing Technologies, I applied programming to automate laboratory processes. I modified Arduino firmware that used thermocouple measurements to switch an oven relay and execute a multi-stage curing profile with defined temperature targets and hold times. I also created reusable Python scripts to generate laser-engraver test-card patterns, improving the speed and consistency of sample preparation. These experiences strengthened my interest in embedded control, hardware interfacing, and practical automation.

I am excited about the opportunity to contribute to Mazdis Innovations and to grow further in embedded firmware in a hands-on environment. Please do not hesitate to contact the UBC Co-op Office at 604.822.6995 or coop.interviews@ubc.ca if you feel that I would be a good candidate.

Sincerely,
Brian Tian

Brian Tian (he/him) Electrical Engineering Student

5868 Agronomy Road, Vancouver, BC, V6T0B5 jxtlx98@163.com | 6043123703

linkedin.com/in/lingxiao-tian

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, SystemVerilog, ARM Assembly
Embedded Systems: ESP32, Arduino, PID control, Embedded C
Software Skills: Git, SolidWorks, MATLAB, ROS 2, Altium Designer, LT Spice, MS Office
Hardware Skills: Circuit design, PCB design, Soldering Iron, Lab equipment

EDUCATION

University of British Columbia **Expected Graduation: May 2028**
B.A.Sc. Electrical Engineering, Fourth Year

Jilin University **2016 – 2020**
B.Sc. Geology

TECHNICAL WORK EXPERIENCE

Convergent Manufacturing Technologies, Vancouver, BC **January 2025 – August 2025**
Co-op Engineer

- Streamlined and maintained SOPs for material analytics and routine lab maintenance, improving workflow efficiency and consistency.
- Supported a new laser engraver by establishing safety practices, authoring end-to-end procedures, and creating reusable Python coding templates to accelerate R&D sample throughput.
- Coordinated DSC/DMA testing, schedules, logs, data exports, and cross-team communication to support on-time material characterization.

TECHNICAL PROJECTS

Franka FR3 Pick-and-Place System (Team Project), UBC **January 2026 – April 2026**

- Ported an OpenCV and Intel RealSense image-processing pipeline into a ROS 2 Python node for real-time, 30-FPS ArUco-based cube localization.
- Integrated and tested the vision, motion-planning, and gripper-control nodes, to convert detected cube coordinates into robot-frame pick-and-place targets.
- Configured the physical 7-DOF Franka FR3 arm using ROS2 open-source library and integration package; validated home positioning, workspace offsets, safety heights, and gripper control.

UBC Open Robotics (Design Team), Vancouver, BC **September 2024 – April 2026**
Arm Electrical Design Team Member

- Programmed and debugged an ESP32 in the Arduino IDE, using breadboard prototypes to confirm control behavior and interfaces before finalizing the board design.
- Led early-stage electrical architecture for the arm (component selection, system diagrams, circuit recommendations), de-risking the design through breadboard validation and embedded programming.
- Coordinated cross-functionally with the mechanical team to select/test motors and verify 3D-printed gears, accelerating iteration by aligning electrical requirements with manufacturable parts.
- Prototyped and validated motor control and feedback paths (A4988/TB6600 drivers, encoders, line receiver), alongside PCB design in Altium.

Laser Projector (Team Project), UBC **January 2024 – April 2024**

- Engineered a finite state machine in Embedded C to control various projector functions and implemented an Interrupt Service Routine (ISR) with a PID controller using control theories
- Troubleshooted and soldered circuit and designed a double-layered PCB in Altium

Name: Lingxiao Tian

THE UNIVERSITY OF BRITISH COLUMBIA
Co-op Program
Faculty of Applied Science

Session	Degr/Catg	Year	Term	Course	#	Credits	Description	Grade
2026S	DEGR	4	2	CPSC_V	213	0	Introduction to Computer Systems	
2025W	DEGR	4	2	CPSC_V	210	4	Software Construction	C
2025W	DEGR	4	2	ELEC_V	433	3	Error Control Coding for Communications and Computers	C
2025W	DEGR	4	2	ELEC_V	442	3	Introduction to Robotics	C+
2025W	DEGR	4	1	CPEN_V	355	4	Machine Learning with Engineering Applications	A-
2025W	DEGR	4	1	ELEC_V	401	3	Analog CMOS Integrated Circuit Design	A-
2025W	DEGR	4	1	ELEC_V	481	3	Economic Analysis of Engineering Projects	C
2025W	DEGR	4	1	FNH_V	200	3	Exploring Our Food	A
2025S	DEGR	4	1-2	APSC_V	210	3	Co-operative Education Work Term	P
2024W	DEGR	4	2	APSC_V	110	3	Co-operative Education Work Term I	P
2024W	DEGR	4	1	APSC_V	450	2	Professional Engineering Practice	A-
2024W	DEGR	4	1	CPEN_V	333	4	Software Design for Engineers II	A-
2024W	DEGR	4	1	CPSC_V	110	4	Computation, Programs, and Programming	B
2024W	DEGR	4	1	ELEC_V	301	4	Electronic Circuits	A-
2023W	DEGR	3	2	ELEC_V	342	4	Electro-Mechanical Energy Conversion and Transmission	A
2023W	DEGR	3	2	ELEC_V	391	6	Electrical Engineering Design Studio II	A
2023W	DEGR	3	1-2	APSC_V	107	0	Introduction to Applied Science Co-op	P
2023W	DEGR	3	1	CIVL_V	250	3	Engineering and Sustainable Development	A
2023W	DEGR	3	1	ELEC_V	311	4	Electromagnetic Fields and Waves	A+
2023W	DEGR	3	1	ELEC_V	315	4	Electronic Materials and Devices	B
2023W	DEGR	3	1	ELEC_V	341	4	Systems and Control	B-
2023W	DEGR	3	1	MATH_V	302	3	Introduction to Probability	A
2022W	DEGR	2	2	ELEC_V	202	4	Circuit Analysis II	A+
2022W	DEGR	2	2	ELEC_V	211	2	Engineering Electromagnetics	A+
2022W	DEGR	2	2	ELEC_V	221	4	Signals and Systems	B+
2022W	DEGR	2	2	ELEC_V	281	3	Technical Communication	B+
2022W	DEGR	2	2	ELEC_V	291	6	Electrical Engineering Design Studio I	A-
2022W	DEGR	2	2	MATH_V	264	1	Vector Calculus for Electrical Engineering	A+
2022W	DEGR	2	1	CPEN_V	211	5	Computing Systems I	B
2022W	DEGR	2	1	CPSC_V	259	4	Data Structures and Algorithms for Electrical Engineers	B-

2022W	DEGR	2	1	ELEC_V	201	4	Circuit Analysis I	A+
2022W	DEGR	2	1	MATH_V	253	3	Multivariable Calculus	A+
2022W	DEGR	2	1	MATH_V	256	3	Differential Equations	A-
2021W	DEGR	1	2	APSC_V	101	3	Introduction to Engineering II	A-
2021W	DEGR	1	2	MATH_V	101	3	Integral Calculus with Applications to Physical Sciences and Engineering	A+
2021W	DEGR	1	2	MATH_V	152	3	Linear Systems	A+
2021W	DEGR	1	2	PHYS_V	170	3	Mechanics I	A+
2021W	DEGR	1	1	APSC_V	100	3	Introduction to Engineering I	B
2021W	DEGR	1	1	APSC_V	160	3	Introduction to Computation in Engineering Design	B+
2021W	DEGR	1	1	MATH_V	100	3	Differential Calculus with Applications to Physical Sciences and Engineering	A
2021W	DEGR	1	1	WRDS_V	150	3	Writing and Research in the Disciplines	C+